Andrew Li

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Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science (Honours, Co-op) — *Class of 2031*

Projects

Nebulosity | *TypeScript, MCP, SQLite, Next.js* | nebulosity.ca | Demo Video — Jul. 2026 – Oct. 2026

- Built an MCP server and code index that made Claude Code up to 4.4× faster on questions that span many files, with all 42 answers fully correct versus 29 without it
- Wrote an indexer for 7 languages that maps all 14,849 files and 140,000+ imports in VS Code's codebase into SQLite and updates on every save, shown as a live map in a published VS Code extension
- Evaluated it over 378 Claude Code runs on 3 open-source codebases, checking every answer against independent reference answers, including one it was never tuned for (up to 3.7× faster)

OlympIQ | *Next.js, TypeScript, Firebase, Firestore* | olympiq.ca — May 2026 – Jul. 2026

- Built a scheduling platform for a 60+ member Science Olympics team, replacing 10+ hours of manual spreadsheet work with a custom algorithm that assigns competitors to their preferred events while avoiding time conflicts
- Shipped it solo and piloted it with the executives before the full team adopts it in the 2026–27 season

Suplex | *Python, FastAPI, Shapely, three.js, React* — Hack the Valley 2026

- Led the technical design of a tool that checks any Toronto address against the City's 4 pre-approved multiplex designs and shows the result in 3D
- Built the rules engine, which screened all 495,456 Toronto lots against the zoning by-law and answers a single address in under 50 ms, plus the FastAPI backend and three.js 3D scene

Experience

Software Engineer — Sep. 2026 – Present
Marble Investments — *Waterloo, ON*

- Building a news pipeline for the firm's equity research platform that sources, filters, and merges the latest news for every tracked company, with an LLM flagging industry news likely to affect them

Software Engineer, Ground Station — Sep. 2026 – Present
UW Orbital (University of Waterloo Satellite Design Team) — *Waterloo, ON*

- Developing the Mission Control Centre, the ground station web app operators use to send commands to the team's 3U CubeSat and monitor its telemetry
- Building the operator dashboard, combining satellite status, command history, telemetry, and the latest image

Head Trainer — Feb. 2023 – Jun. 2026
Science Olympics Team, Victoria Park C.I. — *Toronto, ON*

- Led 6 trainers for a 60+ member team, and all 4 junior teams I coached placed 1st at Western University's Science Olympics

Awards

Hack the North 2026 (349 projects): Best Mobile Experience (Expo), $1,000+ in team prizes
Hack Canada 2025 (89 projects): Best Use of Gemini AI
Canadian Computing Competition (Senior): Distinction, 2024, 2025, 2026
Competitive Programming: DMOJ Top 2%, 500+ problems solved
CALICO Informatics Competition (UC Berkeley): Bronze Brick Winner, 2024
St. Lawrence Coding Competition: 1st Place Team, 2024

Technical Skills

Languages: C++, Java, Python, TypeScript, JavaScript
Frameworks: React, Next.js, Tailwind CSS, three.js, Node.js, FastAPI, Spring Boot
Cloud & Tools: Firebase, Supabase, SQLite, Google Cloud, Cloudflare, Vercel, Git, MCP, Claude Code